Provenance-Aware Belief Aggregation in Multi-Agent Communication Under Convention Heterogeneity

Jayant Dubey
IMS Engineering College
India
dubeymaya787@gmail.com

ABSTRACT

The standard move in multi-agent systems is to weight messages by source reliability. We argue this answers the wrong question whenever reliability and communication convention are distinct latent variables. When agents report through *source-specific conventions*, mappings from internal readings to messages that can invert the signal or transform it depending on context, the real difficulty is interpreting the message, and no reliability weight can fix a misinterpretation. We prove two structural limitations for the convention class studied here. First, no aggregation rule restricted to nonnegative source weights can implement the Bayes-optimal decision when a source uses an inverted convention whose Bayes weight outweighs the remaining sources' combined contribution (the hypothesis of Theorem 1); higher reliability only makes the hypothesis easier to satisfy. Second, no rule assigning a single scalar weight per source, of either sign, can implement the Bayes rule when a source's effective sign depends on observable context. We complement these with a consistency result: a Bayesian aggregator that infers each source's latent convention from a support set concentrates on the observational-equivalence class of the truth as evidence accumulates, so its predictive log-odds converge to the oracle's. The lesson is to decode messages rather than weight them. We introduce a finite-grid Bayesian latent-convention aggregator and evaluate it on a benchmark of eight involutive conventions. Bayesian aggregation consistently outperforms the scalar baselines and is competitive with a trained attention calibrator given identical support information, with the clearest margins at high convention stability and in larger groups. In multi-agent consensus it improves group accuracy by up to 38.2 percentage points over naive averaging, finishing within 4.0 points of an oracle. All results are from a synthetic benchmark; we make no claim about multi-agent communication in general.

KEYWORDS

multi-agent systems, source calibration, trust aggregation, latent-variable inference, collective decision-making

1 INTRODUCTION

Consider a deceptively simple sentence: *"The bridge is safe."* Heard from a structural engineer, it licenses crossing. Heard from a five-year-old, it licenses nothing. Heard from an anonymous account, it licenses suspicion. The message is identical; the decision it supports is not. Something about the *source* changes what the message means.

Multi-agent systems meet this problem at scale. Agents exchange beliefs, recommendations, sensor readings, and every receiver must decide how much weight each message deserves. A substantial class of trust and reputation models represents source differences primarily through reliability or trust parameters, formalised as $\hat{y} = \text{sign}(\sum_s w_s X_s)$ with $w_s \geq 0$. Weighted voting, reliability-weighted averaging, trust-weighted fusion. The nonnegative weight is everywhere. We argue that reliability weighting is insufficient for this setting.

The failure is in how messages are interpreted. Take a source that is perfectly reliable, its internal reading always matching the truth, but which reports through a private convention: it flips its reading, or XORs it with an observable context bit. Its messages are uninterpretable without the convention, and no nonnegative weight, however carefully estimated, repairs the aggregation. Reliability and convention are orthogonal. A source can be maximally trustworthy and still unreadable. We call the resulting problem *convention heterogeneity*. It is distinct from the reliability heterogeneity the literature has studied.

By *provenance-aware* aggregation we mean rules that condition on source identity. The decision-theoretic value of provenance is the change in decision value from knowing who sent the message: with the source known, the receiver can decode through that source's convention; without it, the message may be uninterpretable. Reliability is not convention. Our impossibility theorems characterise regimes where source identity carries this decision value yet no scalar reliability rule can extract it, and our aggregator is a constructive method for extracting it.

Our contributions are fivefold.

(1) We formalise the problem as a structural causal model in which source identity is part of the message-generating process, with involutive conventions, a stability parameter, and reliability kept strictly separate.
(2) We prove that nonnegative source weights cannot implement the Bayes-optimal rule under inverted conventions whenever the inverted source's weight dominates the rest (the hypothesis of Theorem 1).
(3) We prove that no context-independent scalar weight, of either sign, can implement the Bayes rule under context-dependent conventions (Theorem 2).
(4) We establish a consistency result: a Bayesian aggregator that infers each source's latent convention concentrates on the observational-equivalence class of the truth as evidence accumulates (Theorem 3), with

a proof of the equal-reliability characterisation used along the way.

(5) We introduce a finite-grid Bayesian latent-convention aggregator that infers each source's convention, stability, and reliability from a labelled support set, and we evaluate it on a benchmark of eight involutive conventions and nine experiments.

The Bayesian aggregator consistently outperforms the scalar baselines under the synthetic benchmark, with the clearest margins at high convention stability and in larger groups, and is competitive with a trained attention calibrator given identical support information. (The learned calibrator has a slight edge at $N = 3$, which reverses by $N = 8$; see §6.6.)

One clarification before we proceed. The benchmark is synthetic, the convention family finite, the statistics exploratory over ten seeds. Our claim is conditional but sharp: reliability and communication conventions are distinct latent variables, and aggregation methods that model only reliability cannot recover Bayes-optimal decisions when the convention changes what a message means. Reliability is not convention. Weighting evidence is not decoding evidence.

2 RELATED WORK

2.1 Trust and reputation in multi-agent systems

The dominant paradigm for handling source differences is trust and reputation, and it is almost entirely about reliability. TRAVOS [15] estimates the probability that a source provides accurate information; the Beta reputation system maintains Bayesian posteriors over trustworthiness; FIRE, Regret, and BLADE extend the idea to witness reputation and roles. Every one of these models *how often* a source is right, a scalar reliability parameter. That is the gap we work in: a source can be maximally reliable under every one of these frameworks and still be unreadable, because reliability and convention are different things.

2.2 Opinion dynamics and social learning

Classical opinion dynamics, DeGroot [7] and Friedkin–Johnsen [10], aggregate through nonnegative influence weights: exactly the class our Theorem 1 rules out for inverted conventions. Bayesian social learning [1] asks when networks converge to the truth, but takes the mapping from state to signal as fixed and common knowledge. We study what happens when that mapping is private, heterogeneous, and has to be inferred.

2.3 Truth discovery

Truth discovery methods [8, 13, 19] jointly estimate latent truth and source reliability from conflicting claims. Each source still gets a scalar weight, and the weights still determine the aggregation. Convention heterogeneity breaks this parameterisation from the inside: the correct "weight" of a source is not a scalar at all but a context-dependent decoding of the message. Truth discovery asks who to trust. We are asking what they meant.

2.4 Robust and adversarial aggregation

Byzantine-tolerant aggregation [4] and defences against adversarial federated learning [18] handle sources that actively try to corrupt the outcome. Our setting is different: nobody is attacking anyone. The sources are honest and often highly reliable; they simply speak in different conventions, so there is no attacker to detect. What remains is a decoding problem, and reliability scores cannot see it.

2.5 Information theory and pragmatics

The framing grew out of partial information decomposition [3, 17]: does source identity carry information beyond message content? What survived is decision-theoretic – our theorems target decision rules directly, since statistically real information ($I(X; S \mid M) > 0$) need not imply decision value. The intuition is old in pragmatics: Gricean implicature [11] treats speaker identity and context as constitutive of meaning; our conventions are a minimal formal version of that insight.

2.6 Multi-agent LLM systems

Recent work on LLM agents has explored debate protocols [9, 12], round-table deliberation [6], multi-agent evaluation [5], and layered aggregation of proposals [16]. These frameworks model source differences through confidence scores, voting, or an aggregator model's judgement, and empirical studies find debate gains sensitive to protocol design and not always robust over simpler baselines [14]. What they generally do not model is the possibility that agents encode readings through source-specific, context-dependent conventions. Our contribution relative to this literature is the impossibility results: Theorems 1 and 2 show that when the mapping from readings to messages itself varies by source and context, the thing to learn is a decoder, not a weight.

3 PROBLEM FORMULATION

The key modelling choice is that *source identity is part of the generative process of the message*, not a label attached to it afterwards: each source encodes its internal reading through a source-specific *convention* before transmitting, so what the receiver observes, and what it means, depends on who sent it and in what context.

3.1 Generative model

Let $Y \in \{0, 1\}$ be the binary truth with uniform prior, and let there be N sources. Each source s is described by three parameters, and keeping them separate is the entire point of the model:

- A **convention** $c_s \in \mathcal{C}$: an involutive function $f_{c_s} : \{0, 1\} \times \{0, 1\}^d \to \{0, 1\}$ mapping the source's internal reading and an observable context $a \in \{0, 1\}^d$ to a message. Involution means $f_c(f_c(v, a), a) = v$:

applying the convention twice recovers the reading, which is what makes decoding possible at all.

- A **stability** $\rho_s \in [0,1]$: the probability that the source applies its convention on any given transmission rather than reporting its reading directly.
- A **reliability** $r_s \in (1/2, 1)$: the probability that the source's internal reading equals the truth. This is the only quantity the trust literature estimates.

On each round, source s draws reading $R_s = Y$ with probability r_s (else $1-Y$), transmits $M_s = f_{c_s}(R_s, a)$ with probability ρ_s (else R_s), and the receiver observes the message M_s together with the context a.

We work with the **centered message** $X_s = 2M_s - 1 \in \{-1, +1\}$, so that aggregation is sign-symmetric. Define the log-odds weight $\lambda_s = \log(r_s/(1-r_s)) > 0$ and, for inversion-only conventions, the convention sign $\sigma_s = +1$ for honest (identity) conventions and $\sigma_s = -1$ for inverted conventions.

The full generative process for one round, with source parameters $\theta = \{(c_s, \rho_s, r_s)\}_{s=1}^N$ fixed, is:

$$Y \sim \text{Bernoulli}(1/2) \tag{1}$$

$$A_s \sim P_A \quad \text{independently per source and round} \tag{2}$$

$$R_s \mid Y \sim \text{Bernoulli}(r_s) \text{ on } \{Y\}, \text{ i.e. } P(R_s = Y \mid Y) = r_s \tag{3}$$

$$B_s \sim \text{Bernoulli}(\rho_s) \tag{4}$$

$$M_s = \begin{cases} f_{c_s}(R_s, A_s), & B_s = 1 \\ R_s, & B_s = 0 \end{cases} \tag{5}$$

The receiver observes (M_s, A_s) for each source. A **support set** of K labelled triples $\{(M_s^{(k)}, A_s^{(k)}, Y^{(k)})\}_{k=1}^K$ per source is available for calibration; test queries are fresh draws from the same process with the same source parameters.

3.2 Context assumptions

The context A_s is a d-bit vector ($d = 2$ in all experiments), drawn i.i.d. from a fixed distribution P_A independently per source, per round, and independently of the truth Y. It is observed without noise, and the support and test phases share the same P_A. No source chooses or influences the context. These assumptions matter for Theorem 2: P_A must put positive mass on the convention's distinguishing inputs.

3.3 Aggregation classes

We compare two classes of aggregation rules. The **nonnegative-weight class**

$$\mathcal{A}_+ = \left\{ \hat{y}_w(X) = \text{sign}\left(\sum_s w_s X_s \right) : w_s \geq 0 \right\} \tag{6}$$

captures reliability weighting: each source contributes its (centered) message scaled by a nonnegative trust weight. The **signed class**

$$\mathcal{A} = \left\{ \hat{y}_v(X) = \text{sign}\left(\sum_s v_s X_s \right) : v_s \in \mathbb{R} \right\} \tag{7}$$

additionally allows negative weights, i.e. flipping a source's contribution. Both classes assign a single scalar per source, independent of context. In both classes, sign(0) is resolved by an independent fair coin, which makes the $\frac{1}{2}$ tie credit in Lemma 1 exact.

Oracle. The oracle knows the ground-truth parameters (c_s, ρ_s, r_s) for every source and applies the Bayes-optimal aggregation rule directly. No learning, no inference. It is a reference upper bound, not a competitor.

4 STRUCTURAL LIMITATIONS

4.1 Nonnegative weights cannot invert a source

THEOREM 1 (NONNEGATIVE-WEIGHT INSUFFICIENCY). *Suppose sources use inversion-only conventions with full stability ($\rho_s = 1$ for all s), and at least one source j uses an inverted convention ($\sigma_j = -1$). Suppose there exists a message vector X_{-j} from the remaining sources such that $\left|\sum_{s \neq j} \sigma_s \lambda_s X_s\right| < \lambda_j$. Then no rule in $\mathcal{A}_+$ agrees with the Bayes rule on all message vectors. Here $\mathcal{A}_+$ is the class of rules $\hat{y}(X) = \text{sign}(\sum_s w_s X_s)$ with $w_s \geq 0$ fixed per source at decision time: the weights may be estimated from support data by any procedure, but they cannot depend on the current message vector or context.*

PROOF. Consider the inversion-only regime, where the Bayes log-odds are $\text{LO}(X) = \sum_s \sigma_s \lambda_s X_s$. Fix X_{-j} satisfying the hypothesis and form the pair $X^{(1)} = (X_{-j}, -1)$, $X^{(2)} = (X_{-j}, +1)$, differing only in source j's message. Writing $S = \sum_{s \neq j} \sigma_s \lambda_s X_s$,

$$\text{LO}(X^{(1)}) = S + \lambda_j > 0 \quad \text{and} \quad \text{LO}(X^{(2)}) = S - \lambda_j < 0,$$

using $\sigma_j = -1$ and $|S| < \lambda_j$. Hence the Bayes decision is strictly *decreasing* in X_j on this pair: $+1$ then -1.

Any rule $U \in \mathcal{A}_+$ computes $U(X) = \text{sign}(\sum_s w_s X_s)$ with $w_s \geq 0$. Moving from $X^{(1)}$ to $X^{(2)}$ changes the score by $2w_j \geq 0$, so U is *non-decreasing* in X_j and cannot reproduce the Bayes flip from $+1$ to -1. Contradiction. □

The argument is a monotonicity obstruction, and it is worth stating plainly what it rules out: the Bayes rule sometimes has to move *against* a source's message, and no nonnegative weighting can ever do that. The hypothesis is sufficient rather than generic. It names concrete configurations where the failure bites, which is what an impossibility result is for. The cleanest witness is the single-source case: with X_{-j} empty, $S = 0 < \lambda_j$ holds trivially, so a lone inverted source already defeats every nonnegative rule.

COROLLARY 1. *In the inversion-only regime with full stability ($\rho_s = 1$), the signed class $\mathcal{A}$ contains the Bayes rule, attained at $v_s = \sigma_s \lambda_s$.*

4.2 Signed scalars cannot handle context-dependence

Allowing negative weights fixes inversion. It does not fix the deeper problem: when a source's convention conditions on

context, the source's effective sign is itself a function of the context, and no single scalar can represent a function.

THEOREM 2 (SCALAR INSUFFICIENCY FOR CONTEXT-DEPENDENT CONVENTIONS). *Let $f_c(\cdot, a)$ be an involution in its message/reading argument. Suppose there exist contexts $a^{(1)} \neq a^{(2)}$ and an observable message m such that $f_c(m, a^{(1)}) \neq f_c(m, a^{(2)})$.*

(a) There exist observations $x^{(1)} = (X_{-j}, X_j, a^{(1)})$ and $x^{(2)} = (X_{-j}, X_j, a^{(2)})$ sharing the same centered message vector $X = (X_{-j}, X_j)$, with $X_j = 2M_j - 1$, on which the Bayes decisions differ.

(b) Consequently, no context-independent function of the message vector alone—in particular, no scalar weight w_j independent of the current context a—can represent the Bayes-optimal contribution of source j across all observable contexts. A context-dependent weight $w_j(a)$ is a different function class, and is not ruled out.

PROOF. See Appendix A. □

REMARK 1. *Part (a) does the real work: it exhibits a concrete convention class that forces an indistinguishability pair, and part (b) is then immediate. Two assumptions deserve to be explicit. First, the receiver* observes *the context a; without that, context-dependent conventions are trivially undecodable. Second, the convention family is* known *to the receiver; the theorem is about representational capacity, not discovery. The takeaway: context-dependence, not sign, is the fundamental obstacle.*

4.3 Equal-reliability optimum

Lemma 1 quantifies the price of the monotonicity obstruction: with p honest and q inverted sources at reliability r and full stability, the Bayes accuracy counts all $p + q$ decoded readings, while the best unsigned rule attains only the honest-only analogue, i.e. it optimally discards the inverted sources entirely (statement, proof, and a numerical illustration are in Appendix A).

4.4 The Bayesian aggregator is consistent

The results above say what *fixed* rules cannot do. The natural question is whether the Bayesian aggregator of Section 5, which learns the convention instead of fixing a weight, actually finds the truth. It does, in the standard sense.

THEOREM 3 (POSTERIOR CONSISTENCY). *Fix a source with true parameters $\theta^* = (c^*, \rho^*, r^*)$ lying on the finite grid $\mathcal{C} \times$ RHOGRID $\times$ RGRID. Let $[\theta^*]$ be its observational-equivalence class: the grid points inducing the same distribution over support triples (M, a, Y). Then, as $K \to \infty$,*

$$P([\theta^*] \mid \mathcal{D}_s) \xrightarrow{a.s.} 1,$$

and the aggregator's predictive log-odds converge almost surely to the oracle log-odds (exact Bayes inference with known θ^*).

PROOF. See Appendix A. □

REMARK 2. *The theorem promises class concentration, not point identification, and the distinction is load-bearing: on our own grid, distinct parameters can be observationally identical. Under an inversion-only convention, (ρ, r) enter the triple distribution only through the agreement probability $p = \rho(1-r)+(1-\rho)r$, so $(\rho, r) = (1.0, 0.55)$ and $(0.6, 0.75)$—both on-grid—induce identical support-triple distributions; under the identity convention, ρ drops out of the likelihood entirely. What the posterior identifies is the triple distribution itself, and that is all the predictive needs: the query likelihood is the same conditional, constant across each equivalence class, so the predictive log-odds converge to the oracle's regardless.*

A caveat on scope: the theorem assumes θ^* lies exactly on the grid, while our simulations draw ρ and r continuously. Off-grid, the posterior concentrates on the KL-projection of the truth onto the grid [2], and the predictive log-odds converge to the log-odds at that projection. The grid is fine enough ($\Delta\rho = 0.1$, $\Delta r = 0.05$) that the projection is close to the truth, which is why Experiment C still traces the predicted trajectory: at $K = 50$ the Bayesian aggregator reaches 0.821 ± 0.010, essentially the expected reliability ceiling $\mathbb{E}[r] = 0.825$ under full convention stability (Section 6).

5 BAYESIAN LATENT-CONVENTION AGGREGATOR

The theorems say aggregation has to decode messages, not merely weight them. So we infer each source's latent parameters from a labelled *support set* $\mathcal{D}_s = \{(M_i, a_i, Y_i)\}_{i=1}^K$ of past (message, context, truth) triples, and aggregate the resulting predictive log-odds.

Priors. Independent uniform priors over the finite convention family $\mathcal{C}$ and over the discretised stability and reliability grids:

$$c \sim \text{Uniform}(\mathcal{C}), \quad \rho \sim \text{Uniform}(\{0.5, 0.6, \ldots, 1.0\}),$$

$$r \sim \text{Uniform}(\{0.55, 0.60, \ldots, 0.95\}).$$

The prior is discrete over exactly the grid the posterior is enumerated on, so there is no continuous-to-discrete mismatch.

Likelihood. For a support example, let $A_i = \mathbb{1}[f_c(M_i, a_i) = Y_i]$ indicate that decoding the message through convention c recovers the truth, and $B_i = \mathbb{1}[M_i = Y_i]$ indicate direct agreement. With $q_i = \rho A_i + (1 - \rho)B_i$ the probability the transmitted symbol matches the truth under (c, ρ),

$$P(\mathcal{D}_s \mid c, \rho, r) = \prod_{i=1}^{K} \big[(1 - r) + (2r - 1)\, q_i\big]. \tag{8}$$

Posterior. The posterior $P(c, \rho, r \mid \mathcal{D}_s)$ is computed by exact enumeration over the finite grid $\rho \in \{0.5, 0.6, \ldots, 1.0\}$ and $r \in \{0.55, 0.60, \ldots, 0.95\}$. This is a numerical grid approximation, not a closed-form posterior, and we make no claim to the contrary.

Prediction. For a query (M_q, a_q), the predictive log-odds are the log of the ratio of posterior-averaged likelihoods:

$$\text{LO} = \log \frac{\sum_{c,\rho,r} P(c, \rho, r \mid \mathcal{D}_s)\, P(M_q \mid Y = 1, c, \rho, r)}{\sum_{c,\rho,r} P(c, \rho, r \mid \mathcal{D}_s)\, P(M_q \mid Y = 0, c, \rho, r)}, \tag{9}$$

and the aggregator predicts $\hat{Y} = 1$ iff LO > 0. Note this is *not* the posterior average of the parameter-conditional log-odds. The two differ by Jensen's inequality, and only the form above is the exact Bayesian predictive. In the multi-agent setting, per-source predictive log-odds are summed, which is exact under the conditional-independence assumption of Section 3; the oracle below is defined there as well.

6 EXPERIMENTS

6.1 Benchmark design

The benchmark uses eight involutive conventions over two context bits (a_1, a_2): identity, NOT, and six XOR-with-context variants (full table in Appendix B). Each method receives a support set of K labelled (message, context, truth) triples per source and is evaluated on held-out queries. All experiments use 10 seeds (0–9); we report mean ± std. The statistics are exploratory: enough to separate stable gaps from noise, not confirmatory tests.

Baselines. *NoSupport* (content-only); *UnsignedScalar* (non-negative weight, testing Theorem 1); *SignedScalar* (signed weight, testing Theorem 2); *ContextConditioned* (Laplace context-matching); *CtxLogReg* (per-source context-dependent logistic weights); *MetaMean* and *MetaAttn* (DeepSets-style neural calibrators); *Bayes* (the finite-grid aggregator of Section 5, with *BayesUniv*/*BayesFam* variants in the K-sweep); *NaiveMean* and *MajorityVote* (uncalibrated group rules); and the *Oracle* (true (c_s, ρ_s, r_s), Bayes-optimal rule).

6.2 Convention stability sweep

We first vary the convention stability ρ from 0.5 (conventions rarely applied) to 1.0 (always applied), with $K = 10$ and $r \in [0.7, 0.95]$ (Table 1).

Table 1: Accuracy vs. convention stability ρ ($K = 10$); mean ± std over 10 seeds.

ρ	Bayes	MetaAttn	MetaMean	NoSupport	Signed	Unsigned
0.5	0.625 ± 0.011	0.617 ± 0.015	0.621 ± 0.012	0.648 ± 0.008	0.616 ± 0.012	0.627 ± 0.009
0.6	0.640 ± 0.010	0.629 ± 0.014	0.633 ± 0.009	0.624 ± 0.010	0.605 ± 0.009	0.610 ± 0.006
0.7	0.659 ± 0.010	0.640 ± 0.014	0.645 ± 0.010	0.583 ± 0.015	0.591 ± 0.011	0.582 ± 0.010
0.8	0.678 ± 0.009	0.648 ± 0.014	0.655 ± 0.010	0.545 ± 0.010	0.583 ± 0.009	0.560 ± 0.009
0.9	0.714 ± 0.011	0.670 ± 0.017	0.682 ± 0.012	0.512 ± 0.008	0.603 ± 0.011	0.558 ± 0.012
1.0	0.755 ± 0.013	0.696 ± 0.028	0.711 ± 0.016	0.482 ± 0.023	0.621 ± 0.013	0.554 ± 0.015

At low ρ the conventions barely distort messages, so raw-message methods hold their own (unsigned weighting reaches 0.627 at $\rho = 0.5$). As $\rho \to 1$ the picture changes: NoSupport collapses toward chance (0.482), unsigned and signed scalars stall, while the Bayesian aggregator climbs monotonically to 0.755 and the neural calibrators partially recover (0.696–0.711). The price of ignoring conventions grows with their stability, exactly as the theory predicts. The per-query expected reliability ceiling under full convention stability is $\mathbb{E}[r] = 0.825$ even for the oracle; the consensus experiment (§6.6) shows how independent sources combine to lift accuracy far above that ceiling.

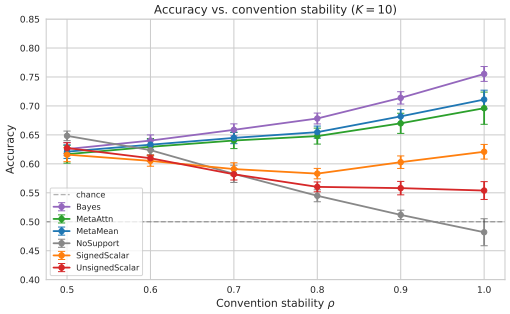


Figure 1: Accuracy vs. convention stability ρ ($K = 10$). Error bars show ±1 std over 10 seeds.

6.3 Representation ablation

At $\rho \approx 1.0$ and $K = 10$, accuracy follows the hierarchy predicted by the theorems (Table 2, Figure 2): content-only near chance, unsigned scalar low, signed scalar at 0.625, context-conditioned scalar at 0.730, the attention calibrator at 0.722, the Bayesian aggregator at 0.758, approaching the oracle at 0.825.

Table 2: Representation ablation at $\rho \approx 1$, $K = 10$; mean ± std over 10 seeds. Methods above the rule use only the support set available to every method; Dec and DecCtx below the rule receive ground-truth decoded features at inference and are diagnostics, not competitors.

Method	Accuracy
Deployable methods (support set only)	
ContentOnly	0.537 ± 0.033
UnsignedScalar	0.550 ± 0.006
SignedScalar	0.625 ± 0.014
ContextConditioned	0.730 ± 0.013
AttentionCalibrator	0.722 ± 0.013
Bayes	0.758 ± 0.015
EMDecoder (MAP point estimate)	0.683 ± 0.013
Oracle	0.825 ± 0.009
Privileged diagnostics (ground-truth decoded features)	
MetaAttn+Dec	0.822 ± 0.008
MetaAttn+DecCtx	0.822 ± 0.008

The table reads as a hierarchy that mirrors the theory: raw messages near chance, unsigned scalars low (Theorem 1), signed scalars at 0.625, context-conditioned scalars at 0.730 (Theorem 2), full latent-convention inference at 0.758, and the oracle at 0.825. Two notes. First, our initial ContextConditioned baseline silently collapsed to SignedScalar; we reimplemented it with Laplace-smoothed context-matching support agreement. Second, the diagnostic MetaAttn+Dec variants receive the true-convention-decoded bit at inference and reach the oracle ceiling (0.822 vs. 0.825), so the learned calibrator's remaining gap is representational, not a capacity problem. Finally, the hand-crafted ContextConditioned (0.730 ± 0.013) has

overlapping 95% CIs with the higher-capacity MetaAttn (0.722 ± 0.013), so the observed difference is not clearly separated at this sample size.

A natural objection is that these baselines are too weak: a sufficiently expressive source × context meta-learner might match Bayes without generative machinery. CtxLogReg tests this: per-source logistic regression on the same $K = 10$ support set, with the logit a context-dependent weight on the message, $w_0 + m\,(v_0 + v_1 a_1 + v_2 a_2 + v_{12} a_1 a_2)$, which nests ContextConditioned's per-context sign and can express any of the eight conventions' decoding maps. It reaches only 0.637 ± 0.007, below ContextConditioned and Bayes with non-overlapping CIs. The result is not a regularisation artefact ($\lambda = 0.01$ and 0.1 agree; accuracy rises with K, $0.63 \to 0.71$ from $K = 10$ to 50). At $K = 10$, expressiveness without the right inductive bias does not help.

A further ablation isolates *why* Bayes beats the scalar baselines, and answers the paper's central empirical question: does posterior uncertainty matter beyond point-estimate decoding? EMDecoder fits the same generative model by EM but decides with the MAP convention and a plug-in log-odds weight: 0.683 ± 0.013, *below* ContextConditioned. Plugging the EM point estimates into the exact oracle log-odds formula recovers 0.732 ± 0.010, matching ContextConditioned but still 2.6 points short of the posterior-predictive 0.758 (non-overlapping 95% CIs). The generative structure alone does not explain Bayes's performance: averaging over convention uncertainty does independent work, and committing to a single estimated convention (MAP recovery only ≈ 0.39 at $K = 10$) throws away a further 4.9 points. Explicit context conditioning recovers much of the lost performance, but Bayesian inference over the latent generative convention provides an additional advantage from uncertainty marginalisation.

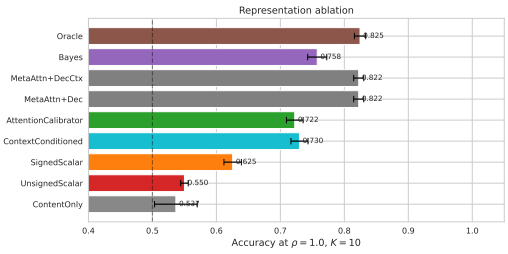


Figure 2: Representation ablation at $\rho \approx 1$, $K = 10$. Error bars show ± 1 std over 10 seeds. Unsigned scalars fail (Theorem 1); signed scalars fix inversion but not context-dependence (Theorem 2).

6.4 Support-size scaling on held-out conventions

Test conventions $\{4, 5, 6, 7\}$; $K \in \{1, 2, 5, 10, 20, 50\}$ (support sets are nonempty, as the neural encoders require). "Held-out" means the following. The neural calibrators are trained on sources using conventions $\{0, 1, 2, 3\}$ and evaluated on sources using conventions $\{4, 5, 6, 7\}$, with K labelled support examples per source available at test time for calibration. BayesUniv considers all eight conventions as candidates. BayesFam restricts its candidate set to $\{4, 5, 6, 7\}$: it knows the test family, and the comparison between the two Bayesian variants therefore measures the value of that knowledge rather than generalisation to unseen conventions.

Table 3: Accuracy on held-out conventions vs. support size K; mean ± std over 10 seeds.

K	BayesUniv	BayesFam	MetaAttn	MetaMean	NoSupport
1	0.588 ± 0.014	0.647 ± 0.019	0.543 ± 0.016	0.523 ± 0.019	0.475 ± 0.032
2	0.644 ± 0.009	0.687 ± 0.015	0.563 ± 0.017	0.559 ± 0.013	0.480 ± 0.035
5	0.711 ± 0.014	0.741 ± 0.014	0.598 ± 0.015	0.578 ± 0.012	0.480 ± 0.039
10	0.767 ± 0.009	0.786 ± 0.010	0.616 ± 0.024	0.589 ± 0.009	0.483 ± 0.034
20	0.793 ± 0.011	0.805 ± 0.010	0.635 ± 0.032	0.603 ± 0.014	0.490 ± 0.032
50	0.820 ± 0.006	0.822 ± 0.006	0.641 ± 0.028	0.621 ± 0.016	0.484 ± 0.029

Both Bayesian variants rise monotonically with K, converging to approximately 0.82 at $K = 50$, which is, up to noise, the information ceiling $\mathbb{E}[r] = 0.825$. The neural calibrators climb from chance to 0.62–0.64 and then flatline, a gap of roughly 18 percentage points at $K = 50$ under the tested conditions. We do not claim this bounds learned calibrators in general; with matched support information and modest capacity, structure beats learning here.

[Figure omitted for space; see Appendix B.]

6.5 Reliability heterogeneity

$\rho \in [0.99, 1.0]$. The three regimes are *mean-matched*: homogeneous $r \in [0.79, 0.81]$, moderate $r \in [0.70, 0.90]$, heterogeneous $r \in [0.65, 0.95]$, all with $\mathbb{E}[r] = 0.80$. This matching matters: an earlier version of this experiment let the heterogeneous regime have lower mean reliability as well, so its performance drop could have been blamed on the level rather than the spread. With the means matched, any remaining difference across regimes is attributable to the *spread* of reliabilities alone.

Table 4: Bayesian robustness to reliability heterogeneity (mean-matched, $\mathbb{E}[r] = 0.80$); mean ± std over 10 seeds.

Regime	Bayes	MetaAttn	NoSupport
Homogeneous	$\mathbf{0.728 \pm 0.013}$	0.699 ± 0.012	0.596 ± 0.009
Moderate	$\mathbf{0.727 \pm 0.017}$	0.707 ± 0.011	0.589 ± 0.010
Heterogeneous	$\mathbf{0.731 \pm 0.011}$	0.706 ± 0.009	0.593 ± 0.006

Bayes outperforms the baselines evaluated in this experiment across all regimes, and its accuracy is flat at ≈ 0.73 across spreads (all three 95% confidence intervals overlap). An ablation pinning reliability to its mean ($r = 0.80$, convention and stability still inferred) reproduces this flat profile exactly ($\Delta \leq 0.002$ in every regime), so the flatness should not be attributed to per-source reliability inference: it reflects robustness of convention identification to reliability misspecification. (The per-query ceiling is $\mathbb{E}[r] = 0.80$; the remaining gap is the usual cost of inferring parameters from $K = 10$.)

[Figure omitted for space; see Appendix B.]

6.6 Multi-agent consensus

N agents observe a shared binary truth, each with its own (c, ρ, r) sampled i.i.d. as follows: $c_s \sim \text{Uniform}(\mathcal{C})$, $\rho_s \sim \text{Uniform}([0.7, 1.0])$, $r_s \sim \text{Uniform}([0.7, 0.95])$. The sampling distribution does not depend on N; realized diversity increases with N because more i.i.d. draws are combined.

Table 5: Group accuracy vs. number of agents under mixed conventions; mean ± std over 10 seeds.

N	Naive	MetaAttn	Bayes	Oracle
2	0.522 ± 0.064	0.742 ± 0.061	0.760 ± 0.038	0.825 ± 0.062
3	0.528 ± 0.049	0.805 ± 0.066	0.798 ± 0.056	0.888 ± 0.043
4	0.562 ± 0.038	0.832 ± 0.059	0.862 ± 0.050	0.910 ± 0.042
6	0.575 ± 0.062	0.852 ± 0.059	0.883 ± 0.047	0.953 ± 0.017
8	0.590 ± 0.048	0.903 ± 0.027	$\mathbf{0.932 \pm 0.031}$	0.968 ± 0.028
12	0.573 ± 0.061	0.937 ± 0.033	$\mathbf{0.955 \pm 0.019}$	0.995 ± 0.008

Naive averaging sits near chance at every N: its sign *is* the majority vote, so the two coincide and we report a single column. Both calibrated methods rise sharply with N; at $N = 12$, Bayes is within 4.0 points of the oracle and 38.2 points above naive averaging. Bayes and MetaAttn have overlapping confidence intervals at small N (0.805 vs. 0.798 at $N = 3$, std ≈ 0.06), so the observed difference is not clearly separated at this sample size; their point estimates separate from $N = 8$ in Bayes's favour, though the gap remains within noise through $N = 12$. As N grows the naive-to-calibrated gap widens: more agents help only if you can decode them.

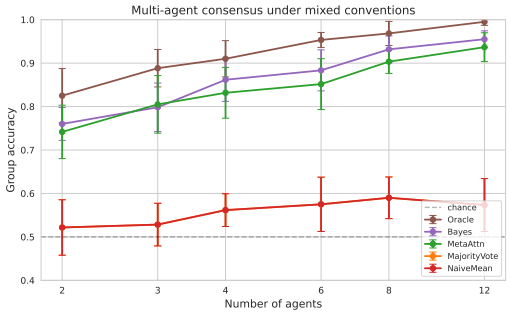


Figure 3: Multi-agent consensus accuracy vs. number of agents. Error bars show ±1 std over 10 seeds.

6.7 Candidate-set misspecification

In practice the modeller never knows the full convention family, so we stress-test a wrong candidate set: sources drawn from all eight conventions while the Bayesian candidate set is restricted to $\{0, 1, 2, 3\}$ (*BayesHalf*); *BayesFull* (all eight) is the well-specified control. $\rho \in [0.99, 1.0]$, $r \in [0.7, 0.95]$, $K = 10$. We report overall accuracy plus the split by whether the true convention was inside the candidate set.

Performance decreases from 0.756 to 0.685 overall, with the reduction concentrated on sources whose true convention lies outside the candidate set. On in-candidate sources the restricted model matches the well-specified control (0.767 vs. 0.756, within noise); on out-of-candidate sources accuracy falls to 0.599, still well above chance, since the posterior spreads mass over the nearest available conventions rather than committing to a single wrong one. Overall BayesHalf drops 7 points (0.685 vs. 0.756). Notably, the learned MetaAttn (0.731) lands between the two Bayesian variants: with no convention family assumed, it handles misspecification better than the restricted Bayesian but worse than the well-specified one. The Bayesian aggregator is a closed-world method, strong inside its assumed family; this experiment bounds the cost of getting the family wrong.

Table 6: Accuracy under candidate-set misspecification; mean ± std over 10 seeds. *BayesHalf* splits accuracy by whether the true convention was inside its candidate set.

Method	Accuracy
BayesFull	0.756 ± 0.013
BayesHalf	0.685 ± 0.012
in-candidate ($c \in \{0..3\}$)	0.767 ± 0.015
out-of-candidate ($c \in \{4..7\}$)	0.599 ± 0.020
MetaAttn	0.731 ± 0.010
Oracle	0.824 ± 0.007

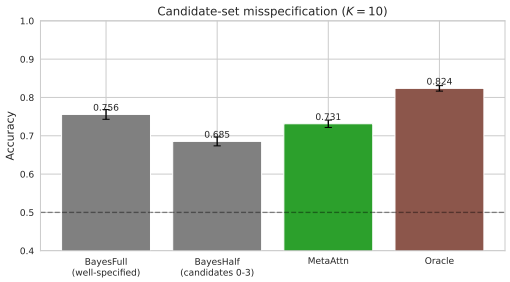


Figure 4: Accuracy under candidate-set misspecification ($K = 10$). Error bars show ±1 std over 10 seeds.

6.8 Correlated sources: breaking the independence assumption

Every result so far keeps the model correctly specified. Here it is deliberately *not* the data-generating process: the Bayesian aggregator assumes sources are conditionally independent given the truth, so we correlate their errors. Ten sources (five pairs) report on one shared truth per query; within-pair query-time errors correlate at ≈ 0.5 (Gaussian copula, calibrated; measured 0.44–0.52), with ExpB draws otherwise ($\rho \in [0.99, 1.0]$, $r \in [0.7, 0.95]$, $K = 10$). Support data stay independent, isolating the misspecification to the combination step. A paired-seed independence arm is the control; MetaAttn could not be re-run here (torch unavailable). Correlation costs every method about a point, *including the oracle*, which also sums independent log-odds: Bayes 0.971 → 0.958, ContextConditioned 0.962 → 0.950, SignedScalar

0.847 → 0.844, naive 0.449 → 0.457, Oracle 0.994 → 0.983 (mean ± 95% CI over 10 seeds; paired per-seed deltas −0.013, −0.012, −0.004, +0.008, −0.011, all within noise of each other). The problem itself gets harder under correlation, and Bayes is not disproportionately hurt: the ranking Bayes > ContextConditioned > SignedScalar > naive is preserved in both arms.

7 DISCUSSION

7.1 Three claims

Nonnegative source weighting is insufficient for the convention class studied here. Theorem 1 proves it; the ablation in §6.3 shows it. Anything that multiplies messages by nonnegative scalars inherits the failure, no matter how the scalars are estimated.

Signed scalars are necessary but not sufficient once conventions depend on context. Theorem 2 makes this precise: the effective sign of a source's contribution is a function of context, and no context-independent scalar can represent a function. The signed-to-context-conditioned gap in the ablation is the empirical fingerprint.

Explicit context conditioning recovers much of the lost performance, but Bayesian inference over the latent generative convention provides an additional advantage from uncertainty marginalisation. ContextConditioned reaches 0.730 against Bayes's 0.758, and the EM ablation shows the gap decomposes into 2.6 points from point-estimate versus posterior-averaging and 4.9 points from hard decoding. Theorem 3 shows the posterior concentrates on the observational-equivalence class of the truth as $K \to \infty$; Experiment C traces that trajectory empirically. The diagnostic MetaAttn+Dec, given privileged decoded features, reaches 0.822 against Bayes's 0.758, so the learned models' bottleneck is representational. Two further stress tests sharpen the claim. First, a discriminative source × context meta-learner (CtxLogReg, §6.3) with the same support information reaches only 0.637, below both ContextConditioned and Bayes: expressiveness without the right inductive bias does not help at $K = 10$. Second, Experiment H breaks the model's independence assumption with correlated source errors and finds Bayes hurt no more than the baselines or the oracle. With matched support information and modest capacity, building in the generative structure beats learning it from scratch.

7.2 What the results do not say

The benchmark is synthetic: a controlled binary task, involutive conventions, uniform priors, deterministic contexts. Real systems have continuous messages, non-involutive mappings, and structured contexts. None of our theorems depend on binarity in an essential way, but the experiments do not test beyond it. Correlated sources are now tested (Experiment H finds no disproportionate harm), but the correlation form is one convenient choice, not an exhaustive study.

The convention family is finite, and the Bayesian method assumes the candidate set is known. Open-world systems would need to grow the set adaptively. Experiment G probes the incomplete-family case and finds smooth rather than catastrophic degradation, but a principled open-world extension is future work, not this paper.

Posterior concentration is moderate at small K and sharpens with support size, just as Theorem 3 predicts.

7.3 Design implications

Two practical lessons. First, *do not aggregate raw messages across heterogeneous sources*: averaging messages from agents with different conventions can do worse than chance. Second, *calibrate before aggregating*: keep a support set per source and combine calibrated log-odds rather than raw messages. Reliability heterogeneity is about *how often* a source is right; convention heterogeneity is about *how* it maps readings to messages. A source can be perfectly reliable and completely unreadable.

8 LIMITATIONS

Messages and truth are binary; the multi-class extension is not worked out, though nothing in the theorems looks essentially binary. The convention family is finite and fixed, and the aggregator assumes the candidate set is known; Experiment G starts on misspecification but an adaptive, open-world candidate set is future work. Sources were assumed conditionally independent given the truth; Experiment H breaks that assumption with correlated errors and finds the ranking preserved, but only one correlation structure was tested. Conventions are static; drifting codes would need online inference, which we do not attempt.

The evaluation is synthetic throughout: no LLM debate, robot teams, or sensor fusion, and the benchmark is not a substitute. The statistics are exploratory over ten seeds—enough to tell stable gaps from noise, not enough for confirmatory claims.

On the method: the grid is coarse (ρ in steps of 0.1, r in steps of 0.05), though a resolution sweep moves every headline number by at most 0.004 (Appendix B.2.1); the priors are flat, and posterior concentration at small K is moderate. The claim is about decoding versus weighting, not optimal grid design.

9 CONCLUSION

Scalar reliability weighting cannot implement the Bayes rule in two regimes: nonnegative weighting cannot invert a source (Theorem 1), and no context-independent scalar, of either sign, can represent a context-dependent convention (Theorem 2). The way out is to decode: our finite-grid Bayesian latent-convention aggregator outperforms the scalar baselines, is competitive with the learned calibrators we tested, and reaches within 4.0 points of the oracle at $N = 12$—up to 38 points over naive averaging. It survives candidate-set misspecification gracefully and is not disproportionately hurt by correlated source errors. Source identity is not a reliability parameter bolted onto a message; it is part of the evidence-to-decision mapping.

REFERENCES

[1] Daron Acemoglu, Munther A Dahleh, Ilan Lobel, and Asuman Ozdaglar. 2011. Bayesian learning in social networks. *Review of Economic Studies* 78, 4 (2011), 1201–1236.

[2] Robert H Berk. 1966. Limiting behavior of posterior distributions when the model is incorrect. *The Annals of Mathematical Statistics* 37, 1 (1966), 51–58.

[3] Nils Bertschinger, Johannes Rauh, Eckehard Olbrich, Jürgen Jost, and Nihat Ay. 2014. Quantifying unique information. *Entropy* 16, 4 (2014), 2161–2183.

[4] Peva Blanchard, El Mahdi El Mhamdi, Rachid Guerraoui, and Julien Stainer. 2017. Machine learning with adversaries: Byzantine tolerant gradient descent. In *Advances in Neural Information Processing Systems*.

[5] Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. 2024. ChatEval: Towards better LLM-based evaluators through multi-agent debate. In *International Conference on Learning Representations*.

[6] Justin Chih-Yao Chen, Swarnadeep Saha, and Mohit Bansal. 2024. ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 7066–7085.

[7] Morris H DeGroot. 1974. Reaching a consensus. *J. Amer. Statist. Assoc.* 69, 345 (1974), 118–121.

[8] Xin Luna Dong, Laure Berti-Équille, and Divesh Srivastava. 2009. Truth discovery and copying detection in a dynamic world. In *Proceedings of the VLDB Endowment*.

[9] Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. 2024. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning*. 11733–11763.

[10] Noah E Friedkin and Eugene C Johnsen. 1999. Social influence networks and opinion change. *Advances in Group Processes* 16 (1999), 1–29.

[11] H Paul Grice. 1975. Logic and conversation. In *Syntax and Semantics 3: Speech Acts*, Peter Cole and Jerry L Morgan (Eds.). Academic Press.

[12] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Zhaopeng Tu, and Shuming Shi. 2024. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 17889–17904.

[13] Jeff Pasternack and Dan Roth. 2011. Making better informed trust decisions with generalized fact-finding. In *Proceedings of IJCAI*.

[14] Andries P Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D Barrett, and Arnu Pretorius. 2024. Should we be going MAD? A look at multi-agent debate strategies for LLMs. In *International Conference on Machine Learning*.

[15] W T Luke Teacy, Jigar Patel, Nicholas R Jennings, and Michael Luck. 2006. TRAVOS: Trust and reputation in the context of inaccurate information sources. In *Proceedings of AAMAS*.

[16] Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, and James Zou. 2024. Mixture-of-Agents enhances large language model capabilities. *arXiv preprint arXiv:2406.04692* (2024).

[17] Paul L Williams and Randall D Beer. 2010. Nonnegative decomposition of multivariate information. *arXiv:1004.2515* (2010).

[18] Dong Yin, Bin Chen, and Rong Pan. 2018. Defense against adversarial attacks in federated learning. In *IEEE International Conference on Big Data*.

[19] Xiaoxin Yin, Jiawei Han, and Philip S Yu. 2008. Truth discovery with multiple conflicting information providers on the web. *IEEE Transactions on Knowledge and Data Engineering* 20, 6 (2008), 796–808.

A DEFERRED PROOFS

A.1 Equal-reliability optimum

LEMMA 1. *For p honest and q inverted sources, all with reliability $r > 1/2$ and full stability ($\rho = 1$), the Bayes accuracy is*

$$A_{\text{Bayes}} = P\big(\text{Bin}(p+q,r) > \tfrac{p+q}{2}\big) + \tfrac{1}{2}P\big(\text{Bin}(p+q,r) = \tfrac{p+q}{2}\big),$$

while the best unsigned (nonnegative-weight) rule attains only

$$A_{\text{unsigned}} = P\big(\text{Bin}(p,r) > \tfrac{p}{2}\big) + \tfrac{1}{2}P\big(\text{Bin}(p,r) = \tfrac{p}{2}\big),$$

i.e. it optimally discards the inverted sources ($w_j = 0$).

Numerically, for $p = 2$, $q = 1$, $r = 0.8$: $A_{\text{Bayes}} = 0.896$ versus $A_{\text{unsigned}} = 0.80$; for $p = 3$, $q = 2$, $r = 0.8$: 0.942 versus 0.896. So nonnegative weighting does not merely underperform. Its optimal strategy is to throw away informative sources entirely. Thus, under this regime, the optimal unsigned rule discards the inverted sources rather than exploiting their information.

PROOF. Work in $\{\pm 1\}$ encoding. With $\rho = 1$, honest sources emit $X_s = Z_s$ and inverted sources emit $X_{p+j} = -Z_{p+j}$, where the readings Z_i are conditionally independent given Y with $P(Z_i = Y) = r$. An unsigned rule computes $\text{sign}\big(\sum_{s=1}^{p} w_s Z_s + \sum_{j=1}^{q} v_j Z'_{p+j}\big)$ with $w_s, v_j \geq 0$, where $Z'_{p+j} = -Z_{p+j}$ satisfies $P(Z'_{p+j} = Y) = 1 - r < \frac{1}{2}$: each inverted message is *negatively* informative about Y.

Step 1: negative information is worse than noise. Fix an inverted source j and write the score as $v_j Z'_j + T$, where T collects all other terms. Conditional on $(T, Y) = (t_0, y)$,

$$P(\text{correct} \mid t_0, y) = (1-r)\mathbb{1}[\text{sign}(v_j y + t_0) = y] + r\mathbb{1}[\text{sign}(-v_j y + t_0) = y],$$

using $P(Z'_j = y \mid Y = y) = 1 - r$ and conditional independence of Z'_j from (T, Y) given Y. Replace Z'_j by a fair coin C_j independent of everything: the conditional accuracy becomes $\frac{1}{2}\mathbb{1}[\text{sign}(v_j y + t_0) = y] + \frac{1}{2}\mathbb{1}[\text{sign}(-v_j y + t_0) = y]$. The difference (coin minus Z'_j) is

$$(r - \tfrac{1}{2})\Big(\mathbb{1}[t_0 y > -v_j] - \mathbb{1}[t_0 y > v_j]\Big) \geq 0,$$

since $r > \frac{1}{2}$ and $-v_j \leq v_j$ (ties broken symmetrically preserve monotonicity). Hence replacing each inverted source's message by pure noise *weakly improves* every unsigned rule. Iterating over $j = 1, \ldots, q$,

$$A(w, v) \leq P\left(\text{sign}\left(\sum_{s=1}^{p} w_s Z_s + \sum_{j=1}^{q} v_j C_j\right) = Y\right),$$

with C_j i.i.d. fair coins independent of (Z, Y).

Step 2: noise cannot beat the honest-only Bayes rule. Write $S = \sum_{s=1}^{p} w_s Z_s$ and $N = \sum_{j=1}^{q} v_j C_j$. Since N is independent of (S, Y) and symmetric,

$$P(\text{sign}(S + N) = Y) = \mathbb{E}_N\big[P(\text{sign}(S + n) = Y)\big].$$

For each fixed n, $\text{sign}(S + n)$ is a deterministic function of the honest readings alone, so its accuracy is bounded by the honest-only Bayes accuracy: the majority vote over decoded readings, $P(\text{Bin}(p,r) > p/2) + \frac{1}{2}P(\text{Bin}(p,r) = p/2)$. Thus $A(w, v) \leq A_{\text{unsigned}}$ for all $w, v \geq 0$.

Step 3: achievability. Setting $v_j = 0$ and equal $w_s > 0$ gives exactly the honest-only majority vote, attaining the bound. The Bayes formula follows identically with all $p + q$ decoded readings. □

A.2 Proof of Theorem 2

Proof. (a) Consider source j in isolation. Because f_{c_j} is an involution in its first argument, the observed raw message M_j decodes to the reading $d^{(t)} = f_{c_j}(M_j, a^{(t)})$ (centered: $X_j = 2M_j - 1$). Context-dependence gives $d^{(1)} \neq d^{(2)}$ for suitable M_j (e.g. the XOR convention with $M_j = v_j \oplus a^{(1)}$). With stability $\rho = 1$, the per-source log-odds are $+\lambda_j$ when the decoded reading is 1 and $-\lambda_j$ when it is 0, so the Bayes decision is $+1$ on $x^{(1)}$ and -1 on $x^{(2)}$ (or vice versa) while the message vector X is identical.

(b) Any context-independent rule is a function $f(X)$ of the message vector alone, so it assigns identical predictions to $x^{(1)}$ and $x^{(2)}$, contradicting (a). □

A.3 Proof of Theorem 3

Proof. Support triples (M_i, a_i, Y_i) are i.i.d. across i. For any grid point $\theta \notin [\theta^*]$, the mean log-likelihood ratio

$$\frac{1}{K}\sum_{i=1}^{K} \log \frac{P(M_i, a_i, Y_i \mid \theta^*)}{P(M_i, a_i, Y_i \mid \theta)} \xrightarrow{a.s.} D_{\mathrm{KL}}(P_{\theta^*} \,\|\, P_\theta) > 0$$

by the strong law of large numbers, with strict positivity from the definition of the equivalence class. The grid is finite, hence so is the set of equivalence classes, and the minimum KL divergence to any class other than $[\theta^*]$ is a positive constant $\delta > 0$. The posterior odds of each foreign class against $[\theta^*]$ therefore decay as $\exp(-K(\delta + o(1)))$, uniformly over the finitely many classes, so $P([\theta^*] \mid \mathcal{D}_s) \to 1$ a.s.

For the predictive, write the log-odds at query (M_q, a_q) as $\log(N_K/D_K)$ with $N_K = \sum_\theta w_\theta\, p_1(\theta)$, $D_K = \sum_\theta w_\theta\, p_0(\theta)$, $w_\theta = P(\theta \mid \mathcal{D}_s)$, and $p_y(\theta) = P(M_q \mid Y = y, a_q, \theta)$. The query likelihood is the same conditional that appears in the support-triple distribution, so $p_y(\theta) = p_y(\theta^*)$ for every $\theta \in [\theta^*]$, for P_A-almost every query context a_q. Hence $N_K = p_1(\theta^*)P([\theta^*] \mid \mathcal{D}_s) + R_K$ with $|R_K| \le \max_\theta p_1(\theta)\,(1 - P([\theta^*] \mid \mathcal{D}_s)) \to 0$ a.s., and likewise for D_K. On the grid, $p_y(\theta) \in [0.05, 0.95]$, so the logarithm is continuous at the limit point and the predictive log-odds converge a.s. to $\log(p_1(\theta^*)/p_0(\theta^*))$: the oracle's log-odds. □

B DEFERRED EXPERIMENTAL DETAILS

B.1 Posterior diagnostic

To check that the Bayesian inference actually identifies conventions rather than stumbling into good predictions by accident, we measure top-1 accuracy on the true convention (0.828 ± 0.040) and posterior mass concentration (0.722 ± 0.032 of sources put mass above 0.50 on the true convention) across 200 random sources with $K = 20$. Decision accuracy is 0.831 when the top-1 convention is correct but still 0.686 when it is wrong: the posterior-predictive averages over convention uncertainty rather than committing to a single convention, so decisions often survive misidentification. Full details are in Appendix B.

Table 7: The eight involutive conventions in the benchmark.

ID	Convention	Involutive	Context-dependent
0	identity	yes	no
1	NOT	yes	no
2	XOR with a_1	yes	yes
3	XOR with a_2	yes	yes
4	XOR with $(a_1 \wedge a_2)$	yes	yes
5	XOR with $(a_1 \vee a_2)$	yes	yes
6	XOR with $(a_1 \oplus a_2)$	yes	yes
7	XOR with $\neg(a_1 \wedge a_2)$	yes	yes

B.1.1 Detailed diagnostic. To check that the Bayesian inference actually identifies conventions rather than stumbling into good predictions by accident, we measure top-1 accuracy on the true convention and posterior mass concentration across 200 random sources with $K = 20$ (mean ± std over 10 seeds):

- Top-1 accuracy on the true convention: 0.828 ± 0.040.
- Fraction of sources with posterior mass exceeding 0.50 on the true convention: 0.722 ± 0.032.

These are diagnostics, not bounds. A diffuse posterior can still predict well if its errors cancel, and a concentrated posterior can predict badly if r or ρ are misestimated. What the numbers say is modest and reassuring: the posterior usually finds the right convention, and the predictive accuracy in §6.1 and §6.5 is consistent with that.

A natural follow-up is whether decisions survive convention misidentification. Conditioning on recovery across the same 200 sources per seed: decision accuracy is 0.831 when the top-1 convention is correct, but still 0.686 when it is wrong, well above chance. The posterior-predictive averages over the convention uncertainty rather than committing to a single (possibly wrong) convention, so the aggregator often gets the decision right even when it cannot name the code. Decision accuracy, not convention-identification accuracy, is what matters, and the Bayesian rule maintains decision accuracy on the former well above chance even when the latter fails.

B.2 Robustness checks

Tables report mean ± std over ten seeds. 95% confidence intervals ($t_{0.975,9} = 2.262$) confirm every headline gap: the ρ-sweep monotonicity, the ablation ordering, the +38.2 point consensus margin and 4.0 point oracle gap at $N = 12$, and the 7 point misspecification drop. Grid-resolution sweeps (96/432/1496 cells) move every headline number by at most 0.004, and a likelihood-temperature sweep preserves all orderings except at the pathological $T = 5.0$. Details are in Appendix B.2.1.

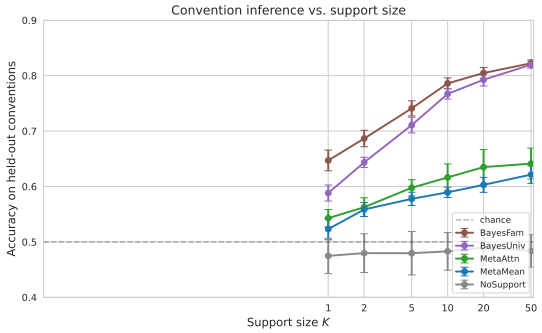


Figure 5: Accuracy on held-out conventions vs. support size K. Error bars show ± 1 std over 10 seeds. Both Bayesian variants converge toward the per-query information ceiling (≈ 0.82); neural calibrators plateau near 0.64.

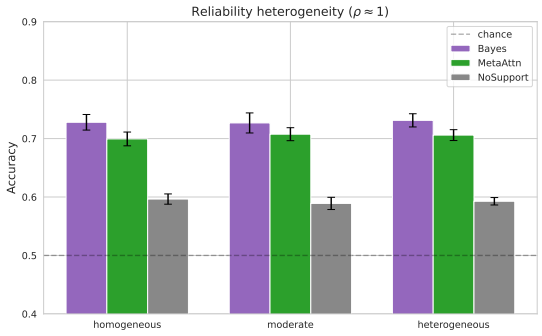


Figure 6: Robustness to reliability heterogeneity ($\rho \approx 1$, mean-matched $\mathbb{E}[r] = 0.80$). Error bars show ± 1 std over 10 seeds. Accuracy tracks the mean reliability across spreads; heterogeneity per se does not degrade the Bayesian aggregator.

B.2.1 Detailed robustness results. Tables report mean $\pm$ std over ten seeds; here we add 95% confidence intervals ($t_{0.975,9} = 2.262$) and two sensitivity sweeps of the Bayesian aggregator, implemented in torch-free NumPy drivers (`code/ci_check.py`, `code/sens_bayes.py`) that reproduce the paper's Bayes columns bit-identically on the default grid. Every headline gap survives: in the ρ sweep Bayes rises monotonically (0.625 ± 0.008 to 0.755 ± 0.009, adjacent intervals non-overlapping) and sits strictly above every baseline at $\rho = 1$; in the ablation Bayes (0.758 ± 0.011) is strictly below the oracle (0.825 ± 0.006) while the privileged MetaAttn+Dec (0.822 ± 0.006) is within noise of it; at $N = 12$ the $+38.2$ point margin over naive averaging (0.955 ± 0.014 vs. 0.573 ± 0.044) and the 4.0 point oracle gap both hold; the misspecification drop (0.756 ± 0.009 to 0.685 ± 0.008) is real. The one comparison that does *not* survive is ContextConditioned vs. MetaAttn (0.730 ± 0.010 vs. 0.722 ± 0.009): overlapping intervals, a point-estimate ordering only, as reported in §6.3. Grid resolution is not a driver: coarsening the grid $4.5\times$ (96 cells) or refining it $3.5\times$ (1496 cells) moves every headline number by at most 0.004. A likelihood-temperature sweep ($T \in \{0.5, 2.0\}$, flattening or sharpening the posterior toward or away from the uniform prior) preserves all orderings; only at the pathological $T = 5.0$ do magnitudes attenuate, and even there Bayes stays above the scalar baselines. The quantitative claims are calibrated to the stated uniform-prior model, and they are not prior-fragile.

REPRODUCIBILITY

Everything is released: the finite-grid Bayesian aggregator, all neural baselines, the smoke tests, and a brute-force unit test (`code/test_predictive_fix.py`) that checks the posterior-predictive log-odds against explicit enumeration. From the repository root, `python code/smoke_test.py` verifies the core components in under ten seconds; `python code/run_all_n10.py` reproduces every table and figure in the paper (ten seeds; a few hours on a CPU-only machine). All randomness goes through a deterministic `SafeRNG` wrapper around Python's `random.Random`; PyTorch seeds are set before every model initialisation. The repository's `REPRODUCIBILITY.md` documents the environment, the per-experiment design, the baseline information budgets, and every correction applied during independent reproduction—including the posterior-predictive fix described in Section 5, which changed the estimator from a posterior expectation of log-odds to the exact predictive quantity. For double-blind review, the code and data are provided as anonymised supplementary material; the camera-ready version will carry a persistent DOI.